



Operational procedures for the Synthetic Raman Spectrometer 'Caillou' during AMADEE-27 EVA

HARDWARE CHECKLIST

- ☐ Caillou instrument
- ☐ Correct handle installed for the mission
- ☐ Dedicated support box
- ☐ Remote control interface

PRE-EVA PREPARATION (HABITAT)

Step	Action	Notes	Duration	Check
1	Charge the internal battery.	Connect via the USB-C port on the main body.	2h	
2	Select the correct handle.	For AA, Rover, Drone	30s	
3	Slide and lock the handle.	Ensure the mechanical connection is secure.	30s	
4	Place instrument inside the support box.	Ensure the instrument is correctly seated.	30s	

PROCEDURE FOR A ROVER (WITH ROBOTIC ARM)

Step	Action	Notes	Duration	Check
Deployment				



1	Position the rover arm above the support box.	Use the arm camera to align with the instrument handle.	3min	
2	Grip and lift the instrument.	It powers on instantly upon removal from the support box.	30s	
3	Transport to area of interest.			
Targeting				
4	Activate the laser beam.	Remote command via the interface.	10s	
5	Align laser with the sample.	Adjust the rover arm until the laser dot is on the target.	1min	
6	Capture context picture.	Remote command via the interface.	1min	
Measurement				
7	Lower the instrument to the ground.	Move until the skirt makes contact with the ground to ensure darkness.	1min	
8	Trigger measurement and wait for data generation.	Remote command via the interface.	30s	
Recovery				
9	Lift the device back up.	Clear the sample area before moving the rover.	1min	



10	Decision point: additional measurements?	If YES: Move the rover to the next target and return to Step 4. If NO: Proceed to Step 11.		
End of EVA				
11	Drive back to the habitat.	Navigate the rover to the garage/docking area.		
12	Store Caillou back in the support box.	The instrument powers off automatically. Confirm End of Mission.	1min	